# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2001-I-Q04] | Part A (Q1-Q5)

## Answer: 291

1. $C = 75^\circ$, and in $ACT$ the other angles are $30^\circ, 75^\circ$, so $AC = AT = 24$.

2. Drop $CH \perp AB$. Special triangles give $AH = 12, CH = 12\sqrt{3}, BH = 12\sqrt{3}$, so $AB = 12 + 12\sqrt{3}$.

3. The area is $(12\sqrt{3})(12 + 12\sqrt{3})/2 = 216 + 72\sqrt{3}$, giving $291$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-I-Q01] | Part A (Q1-Q5)

## Answer: 084

1. Pythagoras in $ABC$ gives $AC^2 = 18^2 + 21^2 = 765$. In $ACD$, $AD^2 = 765 + 14^2 = 961$, so $AD = 31$.

2. Perimeter $18 + 21 + 14 + 31 = 84$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q03] | Part A (Q1-Q5)

## Answer: 578

1. The $5, 12, 13$ triangles $AEB, CFD$ are right at $E, F$. Extend $EA$ past $A$ and $FD$ past $D$ to meet at $G$. The complementary acute angles show $AGD$ is right and congruent to $BEA$, so $GA = 5, GD = 12$.

2. Thus $GE = 17, GF = 17$, with $GE \perp GF$. Pythagoras gives $EF^2 = 2 \cdot 17^2 = 578$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-II-Q05] | Part A (Q1-Q5)

## Answer: 020

1. Scale the original side to six, making each segment of $BC$ length two. The cosine rule gives $AD = AE = 2\sqrt{7}$.

2. Triangle $ADE$ has one-third of the original area, $3\sqrt{3}$. It also has area $\frac{1}{2}AD\,AE\sin\angle DAE = 14\sin\angle DAE$.

3. The sine is $3\sqrt{3}/14$, giving $20$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2014-I-Q01] | Part A (Q1-Q5)

## Answer: 790

1. One level rises $80/3$ mm. Each of the six diagonal crossings has minimum length $\sqrt{50^2 + (80/3)^2} = 170/3$ mm.

2. Adding the bottom width and two ends gives $50 + 6(170/3) + 400 = 790$ mm. Straight taut segments attain the bound.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2015-II-Q04] | Part A (Q1-Q5)

## Answer: 018

1. Let $u = \log 2 > 0$. The altitude is $4u$, while half the base difference is $(\log 192 - \log 3)/2 = 3u$.

2. Each equal leg is the hypotenuse of a $3u$ by $4u$ right triangle, so it is $5u$. The perimeter is $\log 3 + \log 192 + 10 \log 2 = \log(2^{16}3^2)$. Therefore $p + q = 18$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-I-Q04] | Part A (Q1-Q5)

## Answer: 289

1. The original cosine rule gives $\cos A = 7/25$. Put the common length $x$, making $AE = 10 - x > 0$.

2. The cosine rule in $ADE$ gives $x^2 = x^2 + (10-x)^2 - (14/25)x(10-x)$. Cancel and divide by $10 - x$, obtaining $10 - x = 14x/25$.

3. Thus $x = 250/39$, in the required interval, giving $289$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2020-I-Q01] | Part A (Q1-Q5)

## Answer: 547

1. Put $\angle A = \alpha$. Equal sides in $AED$ give $\angle ADE = \alpha$; because $EA$ and $EB$ are opposite rays, $\angle DEB = 2\alpha$.

2. Next $ED = DB$ gives $\angle DBE = 2\alpha$ and $\angle EDB = 180^\circ - 4\alpha$. The straight angle along $ADC$ then gives $\angle BDC = 3\alpha$.

3. From $DB = BC$ and $AB = AC$, both base angles of $ABC$ are $3\alpha$. Thus $7\alpha = 180^\circ$, so $\angle B = 540/7$ degrees and the requested sum is $547$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2021-II-Q05] | Part A (Q1-Q5)

## Answer: 736

1. If the included angle is $\theta$, area is $20\sin\theta$. Its obtuse branch supplies exactly one triangle for each $0 < s < 20$.

2. For the acute branch, the third side satisfies $c^2 = 116 - 80\cos\theta$. The only possible obtuse angle is opposite side $10$, requiring $16 + c^2 < 100$, or $\cos\theta > 2/5$. This branch exists as obtuse exactly for $s < 4\sqrt{21}$.

3. Thus uniqueness among obtuse triangles holds for $4\sqrt{21} \le s < 20$; at the lower endpoint the competing triangle is right. Therefore $a^2 + b^2 = 336 + 400 = 736$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2023-II-Q03] | Part A (Q1-Q5)

## Answer: 250

1. Let the common angle be $\omega$ and leg length be $L$. Triangle $APC$ has angles $90^\circ - \omega, \omega, 90^\circ$, so $L = 10/\sin\omega$.

2. In triangle $ABP$, the angles at $A, B, P$ are $\omega, 45^\circ - \omega, 135^\circ$. The sine rule gives $10\sin 135^\circ = L\sin(45^\circ - \omega)$.

3. Substitution reduces this to $\sin\omega = \cos\omega - \sin\omega$, so $\tan\omega = 1/2$ and $\sin^2\omega = 1/5$.

4. Therefore $L^2 = 500$ and the area is $L^2/2 = 250$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q10] | Part B (Q6-Q10)

## Answer: 083

1. Scale $AC = BC = 1$. In $AMC$, angle $M = 150^\circ$, so sine law gives $CM = 2\sin 7^\circ$. Also $\angle MCB = 83^\circ$.

2. Cosine law in $BMC$ yields $BM^2 = 1 + 4\sin^2 7^\circ - 4\sin 7^\circ \cos 83^\circ = 1$.

3. Thus $BM = BC$, and its base angles at $M, C$ coincide. The requested angle is $83^\circ$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-II-Q07] | Part B (Q6-Q10)

## Answer: 400

1. Let side length be $s$ and $\alpha = A/2$. The diagonals are $2s\cos\alpha, 2s\sin\alpha$. Extended sine law gives $12.5 = s/(2\cos\alpha)$ and $25 = s/(2\sin\alpha)$.

2. Hence $\tan\alpha = 1/2$, with $\sin\alpha = 1/\sqrt{5}, \cos\alpha = 2/\sqrt{5}$, and $s = 10\sqrt{5}$.

3. The area is half the diagonal product, $2s^2\sin\alpha\cos\alpha = 400$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-I-Q07] | Part B (Q6-Q10)

## Answer: 150

1. Extend $AD$ and $BC$ to meet at $P$. Triangle $ABP$ is equilateral. Writing $x = AB$ gives $PD = x - 10, PC = x - 8$ and included angle sixty degrees.

2. Cosine rule gives $144 = (x-10)^2 + (x-8)^2 - (x-10)(x-8) = x^2 - 18x + 84$. Thus $x = 9 + \sqrt{141}$, so $p + q = 150$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-I-Q10] | Part B (Q6-Q10)

## Answer: 032

1. Triangle inequality gives $AD \geq ED - EA = 20\sqrt{7}$. In triangle $ACD$, sine rule gives $AD = AC \sin \angle ACD / \sin 60^\circ \leq 20\sqrt{7}$. Both inequalities must be equalities.

2. Thus $AD = 20\sqrt{7}$, $AC \perp CD$, and $D, A, E$ are collinear in that order. Pythagoras gives $CD = 10\sqrt{7}$, hence $DF = CD \cos 60^\circ = 5\sqrt{7}$.

3. Therefore $EF = ED - DF = 25\sqrt{7}$, giving $32$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-I-Q09] | Part B (Q6-Q10)

## Answer: 113

1. The crease is the perpendicular bisector of $AT$, meeting $AB, AC$ at $P, Q$. Put $AP = PT = x$, $AQ = QT = y$.

2. Triangle $BPT$ gives $x^2 = (12 - x)^2 + 81 - 9(12 - x)$, hence $x = 39/5$. Similarly at $C$, $y^2 = (12 - y)^2 + 9 - 3(12 - y)$, giving $y = 39/7$. Both are between zero and twelve, so the crease endpoints are on the sides.

3. With included angle sixty degrees, $PQ^2 = x^2 + y^2 - xy = 39^3/35^2$. Thus $PQ = 39\sqrt{39}/35$, and the answer is $113$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2000-I-Q14] | Part C (Q11-Q15)

## Answer: 571

1. Scale the common length to $1$ and let $A = \alpha$. In $APQ$, equal sides give $AQ = 2\cos\alpha$ and $\angle APQ = 180^\circ - 2\alpha$. In $ABC$, sine law gives $AB = 1/(2\sin(\alpha/2))$.

2. Using $AB = AQ + 1$ gives $4\sin(\alpha/2)\cos\alpha + 2\sin(\alpha/2) = 1$, hence $\sin(3\alpha/2) = 1/2$.

3. The possible $\alpha = 20^\circ, 100^\circ$ reduce to $20^\circ$ because $AQ > 0$. Thus $r = 80/140 = 4/7$, and the floor is $571$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q12] | Part C (Q11-Q15)

## Answer: 777

1. Let $AD = u, BC = v$ and the common angle be $\alpha$. Cosine law applied to diagonal $BD$ in both triangles gives $u^2 + 180^2 - 360u\cos\alpha = v^2 + 180^2 - 360v\cos\alpha$.

2. Divide by $u - v \neq 0$ to obtain $\cos\alpha = (u+v)/360 = 280/360 = 7/9$. Thus the floor is $\lfloor 7000/9 \rfloor = 777$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-II-Q14] | Part C (Q11-Q15)

## Answer: 463

1. The cevians $AD$, $AE$ are isogonal. Applying sine-area ratios to both and multiplying cancels the sine factors: $(BD/DC)(BE/EC) = AB^2/AC^2$.

2. Thus $BE/EC = (169/196)(6/9) = 169/294$, so $BE = 15 \cdot 169/(169+294) = 2535/463$. The numerator and denominator are coprime, giving $q = 463$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-II-Q15] | Part C (Q11-Q15)

## Answer: 009

1. The nonnegative radical sums make $x, y, z$ positive and imply strict triangle inequalities. The first equation describes an altitude of length $1/4$ to side $x$, with the two nonnegative radical lengths as base parts. The other altitudes are $1/5, 1/6$.

2. If triangle area is $K$, then $(x, y, z) = (8K, 10K, 12K)$. Heron with semiperimeter $15K$ yields $K^2 = 15 \cdot 7 \cdot 5 \cdot 3\, K^4$, so $K = 1/(15\sqrt{7})$. The ratio triangle is acute, confirming all altitude feet are internal.

3. Thus $x + y + z = 30K = 2/\sqrt{7}$ and $m + n = 9$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q11] | Part C (Q11-Q15)

## Answer: 179

1. At simultaneous ground/small-cylinder tangency, center separation is $96$ and vertical difference $48$, so the center line is $30^\circ$ from horizontal. Crossing sweeps the large center through $120^\circ$ on radius $96$, advancing $2 \cdot 96 \cos 30^\circ = 96\sqrt{3}$.

2. No slip around the stationary small circle makes large-cylinder spin $(96/72)120^\circ = 160^\circ$. The remaining $200^\circ$ spin is on flat ground, advancing $(200/360)2\pi \cdot 72 = 80\pi$.

3. Total displacement $80\pi + 96\sqrt{3}$ gives $179$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-II-Q14] | Part C (Q11-Q15)

## Answer: 007

1. Use rectangle vertices $A=(0,0), B=(a,0), C=(a,b), D=(0,b)$ and points $E=(x,0), F=(a,b-y)$. The three equal quantities are exactly squared sides of equilateral triangle $DEF$.

2. Let $\theta=\angle ADE$. Then $x=b\tan\theta$, $y=a\tan(30^\circ-\theta)$, and equal sides give $a/b=\cos(30^\circ-\theta)/\cos\theta=\sqrt{3}/2+\tan\theta/2$. Nonnegative $x,y$ imply $0\le\theta\le 30^\circ$.

3. The ratio increases with $\theta$ and reaches $2/\sqrt{3}$ at $\theta=30^\circ$, with $y=0, x=b/\sqrt{3}<a$. Thus $\rho^2=4/3$ and the answer is $7$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q12] | Part C (Q11-Q15)

## Answer: 676

1. Write the bases as $8a, 7a$ for positive integer $a$, and common perimeter as $P$. An isosceles triangle with base $b$ and perimeter $P$ has area squared $b^2P(P-2b)/16$.

2. Equal areas yield $64(P-16a) = 49(P-14a)$, hence $P = 338a/15$. The equal legs in the respective triangles are $(P-8a)/2 = 109a/15$ and $(P-7a)/2 = 233a/30$.

3. Because $233$ is coprime to $30$, integrality requires $30 \mid a$. Taking $a = 30$ gives sides $(218, 218, 240)$ and $(233, 233, 210)$, both nondegenerate and of perimeter $676$.

4. Thus the least perimeter is $676$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q14] | Part C (Q11-Q15)

## Answer: 007

1. Let $\alpha = \angle ACP$ and $u = AP$. In triangle $APC$, $\angle A = 180^\circ - 3\alpha$, so $45^\circ < \alpha < 60^\circ$. The sine rule gives $u = \sin\alpha / \sin 3\alpha = 1/(4\cos^2\alpha - 1)$ and $AC = 2u\cos\alpha$.

2. The large right triangle gives $AC = 4\cos A = -4\cos 3\alpha$. Divide the two expressions by $2\cos\alpha$ to obtain $u = 6 - 8\cos^2\alpha$.

3. Substitute $4\cos^2\alpha = 1 + 1/u$, yielding $u^2 - 4u + 2 = 0$. Since $\alpha > 45^\circ$, the first expression for $u$ makes $u > 1$, selecting $u = 2 + \sqrt{2}$.

4. Then $BP = 2 - \sqrt{2}$ and $AP/BP = 3 + 2\sqrt{2}$, giving $7$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-II-Q12] | Part C (Q11-Q15)

## Answer: 112

1. Let the four diagonal portions be $a = AP, b = BP, c = CP, d = DP$. The common sine of the intersection angle cancels from the area equality, giving $(a - c)(b - d) = 0$. Thus at least one diagonal is bisected.

2. Let $u$ be half the bisected diagonal and $v, w$ the other portions, orienting the angle $\theta$ appropriately and putting $v$ next to the fourteen-unit side. Averaging and subtracting the adjacent cosine-law equations gives $u^2 + v^2 = 148, u^2 + w^2 = 180, uv|\cos\theta| = 24, uw|\cos\theta| = 40$. These relations apply whether $a = c$ or $b = d$, with the diagonal names exchanged.

3. Hence $w/v = 5/3$ and $w^2 - v^2 = 32$, so $v = 3\sqrt{2}, w = 5\sqrt{2}, u^2 = 130$. It follows that $\cos^2\theta = 16/65$, and $\sin\theta = 7/\sqrt{65}$.

4. The area is $u(v + w)\sin\theta = \sqrt{130}(8\sqrt{2})(7/\sqrt{65}) = 112$.

Source / 来源

# Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2025-II-Q12] | Part C (Q11-Q15)

## Answer: 019

1. The common sine is $5/13$, so consecutive radial lengths multiply to $26/5$. Hence the ten radial lengths alternate between $a, b$, with $ab = 26/5$, and the desired sum is $u = a + b$.

2. Cosine rule makes each of the nine intervening polygon sides have square $a^2 + b^2 - 2ab(12/13) = u^2 - 20$. Perimeter gives $u + 9\sqrt{u^2 - 20} = 20$.

3. Squaring with $20 - u > 0$ yields $4u^2 + 2u - 101 = 0$, so $u = (9\sqrt{5} - 1)/4$. The requested sum is $9 + 5 + 1 + 4 = 19$.

Source / 来源